# When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models

H M Akshay[1*], Gaurav S B[1], Eshita Narayana[1], Harshita Srivastava[1], Malini M Patil[1]

[1]Department of Computer Science and Engineering, RV Institute of Technology and Management, Bengaluru, 560083, Karnataka, India.

*Corresponding author(s). E-mail(s): rvit23bcs106.rvitm@rvei.edu.in;
Contributing authors: rvit23bcs258.rvitm@rvei.edu.in;
rvit23bcs031.rvitm@rvei.edu.in; rvit23bcs323.rvitm@rvei.edu.in;
malinimp.rvitm@rvei.edu.in;

**Abstract**

**Purpose:** Graph neural networks (GNNs) are proposed for healthcare insurance fraud because collusive provider rings leave traces in claim networks; when they beat tabular models is unclear. **Methods:** We built a synthetic claims benchmark that independently varies the feature-level signal, doctor–patient density and claim volume of fraud rings, and tests without leakage on rings of unseen members. Fourteen equally tuned methods, including gradient boosting with and without engineered graph features and five heterogeneous GNNs, were compared in nine regimes (twelve seeds each), at four ring volumes, and on public provider-labelled Medicare claims. **Results:** At default volume (ring patients filing 5.7 times as many claims as others), per-entity aggregates detected most rings (average precision 0.89–0.92) and graph features added up to 0.10 for dense rings. At four to eight claims per month, aggregates fell to 0.05–0.42, graph features reached 0.41–0.92 for moderately and densely linked rings, and pure GNNs trailed them by 0.17–0.54. Averaged over regimes, no pure GNN outperformed gradient boosting with graph features (upper 95% limits $\leq$0.006), hybrids of both gained 0.01, and GNNs were 0.09–0.14 worse on isolated fraud. Recurring ring members inflated performance by up to 0.60. On the public data, gradient

boosting on provider aggregates was best (average precision 0.704). **Conclusion:** Relational structure helps when collusion forms dense blocks of otherwise unremarkable members, and a few engineered shared-patient features captured it better than message passing alone. Graph models should be evaluated against graph-augmented tabular baselines on unseen entities.

**Keywords:** Healthcare fraud detection, Provider rings, Heterogeneous graph neural networks, Gradient boosting, Synthetic benchmark, Data leakage

# 1 Introduction

Health insurance fraud causes losses estimated in the tens of billions of dollars each year in the United States alone [1]. Detecting it is a routine task of payers' claims-processing and audit systems, which must decide from administrative claims data which providers and claims a limited team of investigators examines. Fraud takes two broad forms. In *isolated* fraud a single provider inflates or fabricates individual claims, so the evidence lies in the claim itself: amounts, procedure counts or coding that are unusual for the provider's specialty. In *collusive* fraud a group of providers and recruited patients file claims together. Each claim can look ordinary; the evidence lies in who bills whom—patients who see many members of the same small group of providers, and providers who share an unusually large fraction of their patients.

The second form motivates graph neural networks (GNNs) over the patient–provider–facility network, in particular heterogeneous architectures such as RGCN [2], HAN [3], HGT [4] and Simple-HGN [5]; GNNs of this kind have also been built specifically for health-insurance claims [6–8]. Whether GNNs are needed, however, is an empirical question. Gradient-boosted trees remain strong on tabular data [9], relational signals can be engineered as features, and benchmark studies outside healthcare report that tree ensembles with neighbourhood features often match or beat GNNs on graph anomaly detection [10, 11]. On public Medicare claims with provider fraud labels, machine-learning models given graph-centrality features outperformed GNNs at a fraction of the training cost [12]. A single dataset of this kind, however, cannot show *when* relational structure helps or *why*: its labels mark providers, not collusive groups, and neither the typology nor the strength of the relational signal can be varied.

Answering the question for healthcare claims is difficult for two reasons. First, public claims data with labelled collusion are not available, and real labels conflate fraud typologies; on the most widely used public provider-labelled dataset, reported results vary widely between studies and are rarely accompanied by significance tests or code [13]. Second, evaluation is fragile: identifiers that encode the label, statistics computed on the full dataset, and providers that appear in both training and test data inflate apparent performance [14, 15]. Fraud rings make the last problem acute, because a model can score a test claim correctly by recognising a provider it saw labelled during training instead of recognising coordination.

We therefore study the question in a controlled setting and check it on public provider-labelled data. Our contributions are:

1. **A map of when relational structure helps.** We vary three properties of collusive rings independently—how anomalous their claims are ($\delta$), how densely their doctors and patients are linked ($s$), and how many claims they file—and compare 14 methods under an equal tuning budget: tabular models with and without engineered graph features, a multilayer perceptron (MLP) control, five heterogeneous GNNs, two hybrids and an unsupervised overlap score. The result is a map of the conditions under which structure adds information beyond per-entity features, and of whether message passing or a few engineered features capture it (Sections 6 and 9).
2. **A reusable, leakage-tested benchmark.** The generator's *emergent* protocol forms every test ring from doctors and patients never seen in a training ring, and its audit-period design keeps features and graphs free of future information; automated tests enforce both (Section 3). Code, generator and tests are publicly available.
3. **Evaluation pitfalls, quantified.** Test rings whose members recur from training inflate ring detection by up to 0.60 average precision and can reorder methods; and pooled tests over generated worlds that share their background data overstate significance, so the generator seed, not the world, must be the unit of replication. A re-analysis of a small fixed benchmark shows how such data can suggest the opposite conclusion (Sections 7 and 9.3).
4. **Mechanism.** Degree-preserving rewiring, ablations and an analysis of which inputs the trained models use show that GNNs detect dense rings by counting shared counterparties, which explains when the engineered features suffice (Section 7).
5. **A check on public data**: a provider-labelled Medicare claims dataset evaluated with a provider-disjoint protocol, equal tuning budgets, repeated cross-validation, per-rotation model selection and tests corrected for overlapping training sets (Section 5). It replicates, under stricter conditions, an earlier finding on public Medicare provider data [12]. Its labels identify providers, not rings, so it tests whether graph information helps provider-level detection, not whether rings are detected.

Our main findings are as follows. Claim features alone cannot reveal rings whose claims look legitimate. When ring members are unusually active, as in our default setting, per-entity aggregates capture most of the evidence; as ring volume falls, aggregates lose most of their power and coordination structure carries most of the evidence, until rings that file about one claim per member are found by no method. Structure adds most when the ring's doctor–patient block is moderately or highly dense. Where it does, gradient boosting with a few engineered shared-patient features was as good as or better than every pure GNN we tested, and far better when ring volume was low; GNNs that also receive those features gained a small but reliable amount in average precision (AP; +0.01 averaged over regimes), most for sparse rings, where a pre-declared test on fresh seeds confirmed it at low volume. GNNs are clearly worse on isolated fraud, because message passing dilutes claim-level anomalies. At default volume HGT's ring detection is structural: it collapses without degree encodings and survives with degree encodings as its only inputs. Routing high-risk claims from gradient boosting to a GNN gained nothing; fusing the two models' rankings gained at most 0.006 AP, as an ensemble rather than as a router. Evaluations that let test rings

share members with training rings overstate ring detection, by up to 0.60 AP for a model that only sees entity attributes. On the public provider-labelled data, where labels show no relational assortativity, no graph-based method improved on gradient boosting over provider aggregates, in line with earlier work on these data [12].

## 2 Related Work

***Healthcare fraud detection.***

Most published healthcare-fraud detectors are supervised classifiers over claim- or provider-level features: classical learners and neural networks on Medicare data labelled with the federal exclusion list (LEIE) [16–18], and multidimensional outlier analysis on Medicaid data [19]. Relational information enters, if at all, through hand-engineered features. On the public provider-labelled dataset used in Section 5, reported F1 scores range from 0.49 to 0.90 under differing protocols, and significance tests and released code are rare [13].

***Graph-based fraud detection.***

Coordination leaves structural traces such as dense blocks, lockstep behaviour and unusual egonets [20, 21], which unsupervised detectors exploit directly [22–24]. GNNs have been applied to review fraud [25], underground-forum key players [26] and camouflaged or class-imbalanced fraudsters [27, 28], and spectral analyses argue that standard message passing can blur anomalies [29]. Heterogeneous GNNs use relation-specific weights (RGCN [2]), attention over hand-chosen meta-paths (HAN [3]) or attention parameterised by node and edge type (HGT [4]); Simple-HGN [5] showed that a GAT with edge-type embeddings is competitive with them when baselines are tuned fairly.

***Graph learning for health-insurance fraud.***

On data from single payers, heterogeneous GNNs have classified patients with meta-path attention [6] or graph structure learning [7], and billed activities with several heterogeneous GNNs and post-hoc explainers [8]; the baselines were other graph models [6] or tabular models without relational features [8]. On the public provider-labelled dataset, a graph transformer network reached an F1 score of 0.53 against 0.52 for a random forest in a study of graph attention networks [13, 30], meta-path embeddings computed on a 10% claim sample matched tree ensembles in ROC-AUC (0.87–0.88) [31], and machine-learning models given graph-centrality features outperformed GNNs in precision, recall and F1 while training 250–300 times faster [12]; our analysis of these data replicates the last finding under a stricter protocol. Because their labels mark providers, patients or activities rather than collusive groups, these studies cannot show under which conditions relational structure carries information that per-entity features miss.

***When do graph models beat tabular models?***

Evidence is mixed. A random forest on node features outperformed graph convolutional networks on the Elliptic Bitcoin data [10]; tree ensembles with simple

neighbourhood aggregation were among the strongest supervised graph anomaly detectors across ten datasets [11]; gradient-boosted trees with graph-based input features were very strong, though not universally winning, baselines on industrial graphs [32]; a graph gradient-boosting machine was on par with or better than the state of the art for insurance fraud [33]; and tree ensembles remain hard to beat on tabular data [9]. Synthetic benchmarks with injected typologies make the question testable, as for money laundering [34] and tabular fraud [35]. We found no comparable controlled study for heterogeneous patient–provider–facility claim graphs, where the relevant typology is the collusive provider ring.

***Evaluation pitfalls.***

Features that encode the target, preprocessing fitted on the full dataset and entities shared between training and test data inflate reported performance [14, 15]. GNN benchmarks are also sensitive to split choice and tuning effort [36], and duplicate nodes that leak between training and test data strongly affected GNN performance on two popular benchmarks [37]. Fraud rings make entity overlap acute: a model can score a test claim correctly by recognising a provider it saw labelled in training rather than by recognising coordination.

# 3 Benchmark Design

## 3.1 Synthetic claims world

We simulate 6,000 patients, 300 doctors and 40 hospitals in five regions over four audit periods of six months (Fig. 1; Table A1). Each doctor has one of eight specialties, which sets the distribution of claim amounts, inpatient probability and procedure counts. Patients have heterogeneous utilisation (gamma-distributed claim rates with mean three claims per two years) and one to three regular doctors in their region; 85% of legitimate claims go to a regular doctor and the rest to a popularity-weighted random doctor. Claim amounts are log-normal given the specialty, procedure count and inpatient status.

***Isolated fraud.***

Three per cent of claims are upcoded claims filed through a patient's own regular doctor. Their log-amount, procedure count, inpatient probability and procedure-code level are shifted by two standard deviations; their structure is that of an ordinary claim.

***Provider rings.***

A ring consists of four colluding doctors and twelve recruited patients, drawn preferentially from high utilisers. During its active period the ring files 16 claims per month between linked doctor–patient pairs, clustered around monthly burst days. Two parameters are varied independently:

- the *feature shift* $\delta \in \{0, 1, 2\}$ applied to ring claims, in standard deviations of log-amount (with matching shifts in procedures, inpatient probability and coding level).

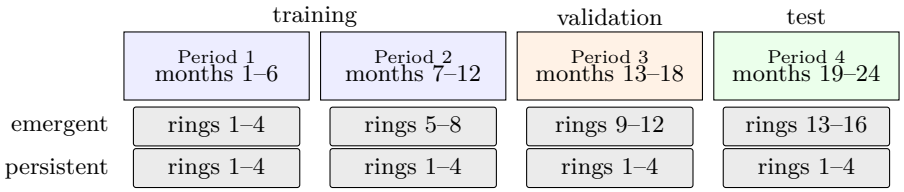



**Fig. 1** Audit-period protocol. Each six-month period is analysed separately: features and graphs use only that period's claims. Under the emergent protocol every period has new rings formed by disjoint doctors and patients, so test rings are unseen; under the persistent protocol the same rings recur and can be recognised by their members.

At $\delta = 0$ a ring claim is drawn from exactly the distribution of a legitimate claim of the same doctor's specialty;

- the *structural density* $s \in \{0.1, 0.5, 0.9\}$, the probability that a ring patient is linked to each ring doctor. Ring claim volume is fixed, so $s$ changes only how the claims are spread over the doctor–patient block (realised block densities $\approx 0.37$, 0.62 and 0.79).

Ring patients keep a single regular doctor, so the number of distinct doctors they see is governed by $s$. Their elevated claim volume, a degree signal visible to every model, is the same in all regimes and large: in a test period a ring patient files on average 9.3 claims, against 1.6 for other patients with a claim (99th percentile about 5), and a ring doctor 38 claims against 16. Section 6.6 reduces ring volume to four claims per month, which brings ring patients to 3.7 claims, about the 95th percentile of other patients, and ring doctors to 20.

***Protocols.***

Under the *emergent* protocol (default), four new rings start in each period and rings in different periods are formed by disjoint doctors and patients, so every test ring is unseen during training. Under the *persistent* protocol the same four rings are active in all periods, which reproduces the setting of most prior synthetic evaluations and lets a model succeed by recognising known members. Both protocols have identical claim and ring-claim volumes per period. Ground truth (fraud type and ring identifier) is stored separately from the claim records and never enters feature construction.

***Replication.***

Each regime is generated from twelve seeds. The generator draws a world's entities, legitimate claims, isolated fraud and recruited ring members before it applies the ring parameters, so for a given seed the nine regimes share all of these and differ only in their ring claims. Regimes are therefore compared on otherwise identical worlds, and analyses pooled over regimes treat the seed, not the regime–seed world, as the unit of replication (Section 4.5).

### 3.2 Evaluation protocol

Periods 1–2 are used for training, period 3 for validation (early stopping, threshold selection) and period 4 for testing. Each period is analysed on its own, as in a retrospective audit: a claim's aggregate and graph features, and the graph a GNN operates on, are built only from claims of the same period. This keeps every split's features on the same time scale and prevents information from later periods reaching earlier ones; it is verified by automated tests (Section 10).

## 4 Methods

### 4.1 Tabular features

- **C** (claim): log-amount, procedures, length of stay, inpatient flag, coding level, month (cyclical), patient age, sex and chronic-condition count, doctor experience and specialty, hospital size.
- **A** (aggregates within the period, no labels): the patient's claim count, distinct doctors and mean/maximum log-amount; the doctor's claim count, distinct patients and mean log-amount; hospital claim count; patient–doctor pair count; and the claim's log-amount relative to the patient, doctor and specialty means.
- **G** (graph features on the period's patient–doctor bipartite graph, no labels): the doctor's maximum shared-patient Jaccard similarity with another doctor and number of doctors sharing at least two patients, the mean degree of the doctor's patients, the patient's mean doctor overlap and share of doctor pairs that co-treat other patients, and the triadic closure of the claim (how many of the patient's other doctors share patients with the claim's doctor), raw and degree-normalised.

### 4.2 Heterogeneous graph

Each period's graph has patient, doctor, hospital and claim nodes and the relations *files* (patient→claim), *billed_by* (claim→doctor), *affiliated* (doctor→hospital) and *visits* (patient→doctor), each with a reverse relation so that every node type receives messages. Claim nodes carry claim-intrinsic attributes; entity nodes carry their static attributes and $\log(1 + \text{degree})$ for each incident relation, a label-free structural encoding included because attention- and mean-based aggregators cannot otherwise count neighbours [38]. No engineered aggregates or labels are used as node features (Fig. 2). Claim attributes and degree encodings are standardised with statistics of the training periods; static entity attributes, which do not change over time and carry no labels, with statistics over all entities.

### 4.3 Models

- **Tabular**: logistic regression (LR) and random forest (RF) on C+A; histogram gradient boosting (GB) [39] on C, C+A and C+A+G.
- **MLP-C**: a multilayer perceptron on C, the no-graph control for the GNNs.
- **GNNs**: heterogeneous GraphSAGE [40] (one convolution per relation, mean or sum aggregation), RGCN [2], HAN [3] (claim nodes; meta-paths claim–patient–claim

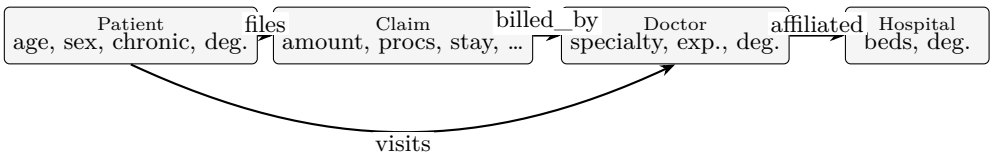



**Fig. 2** Heterogeneous claim graph of one audit period. Every relation also has a reverse relation (not drawn), so every node type receives messages and doctor context reaches claim nodes. Entity nodes carry static attributes and log(1+degree) per incident relation; claim nodes carry only claim-intrinsic attributes.

and claim–doctor–claim, with patient, doctor and hospital attributes appended to claim inputs), Simple-HGN [5] and HGT [4]. All share the same per-type input projection, residual connections with layer normalisation, and two-layer classification head.

- **Hybrids**: SAGE+AG and HGT+AG, whose claim nodes also receive the A and G features, testing whether message passing adds information beyond the engineered features. Hybrids reuse the tuned configuration of their base architecture and receive no separate tuning budget.
- **Overlap score**: the degree-normalised triadic closure of the claim, used directly as an unsupervised score.

Tabular models use scikit-learn [41] and graph models PyTorch Geometric [42]. Neural models are trained full-batch with AdamW, class-weighted binary cross-entropy and early stopping on validation average precision (patience 20, at most 200 epochs). Every model is trained once per world with the same training seed, so variation between worlds includes variation due to initialisation.

***Hyperparameters.***

Every method receives the same budget of up to ten configurations (the first being a default), selected by mean validation average precision on two development worlds (generator seeds 901 and 902, $\delta = 0.5$, $s = 0.5$) that are never used for evaluation (Online Resource 1, Table S1). The development regime ($\delta = 0.5$) is deliberately not one of the evaluation regimes.

## 4.4 Two-stage routing

A common deployment proposal is to score every claim with a cheap tabular model and pass only high-risk claims to a GNN. We evaluate this with GB-CAG as the first stage and HGT or SAGE as the second. Claims whose GB-CAG score exceeds a threshold set on validation data are escalated, with the share of escalated claims chosen by validation AP, and are ordered either by the GNN score alone (*re-rank*) or by the mean of the two models' validation percentiles (*fused*). The stages are the models of the main experiment, retrained with the same seeds; their test predictions reproduce the stored ones exactly for the GNNs and to within $3 \times 10^{-8}$ for GB-CAG (float32 storage). Details are in Online Resource 1, Section S2.

### 4.5 Metrics and statistics

The primary metric is average precision (AP, the area under the precision–recall curve), appropriate under class imbalance [43]. *Ring AP* scores ring claims against legitimate claims (isolated fraud excluded); *isolated AP* is defined analogously. We also report F1 at the threshold that maximises validation F1, the area under the receiver operating characteristic curve (ROC-AUC) and, because investigators can examine only a limited number of cases [44], recall when the top 5% of claims are investigated. For routing we add precision among the top 5% of claims (P@5%) and, as a ring-level measure, the share of test rings with at least one claim in that budget.

Each of the nine regimes is replicated with twelve worlds (generator seeds 1–12), 108 worlds in total. Seeds 1–6 formed a first round; seeds 7–12 were added in a second round whose size, analyses and reporting were fixed in a plan committed to the repository before any of its runs started, after the first round's results had been seen. Within a regime the twelve worlds are independent, and methods are compared with paired $t$-tests over them. Because the nine regimes of a seed share most of their data (Section 3.1), comparisons pooled over regimes average each seed's nine paired differences and test the twelve seed means ($t$-test with eleven degrees of freedom); treating the 108 worlds as independent would overstate the evidence. The persistent-ring, ablation, routing and ring-volume analyses use seeds 1–6. $p$-values are Holm-adjusted within each pre-specified family of comparisons [45], Wilcoxon signed-rank tests are a robustness check [46], and intervals are 95% confidence intervals (CIs) from the $t$-distribution over worlds or, for pooled comparisons, over seeds. A non-significant difference is not evidence that two methods are equivalent; the CIs show which differences remain compatible with the data.

### 4.6 Use of AI tools

An AI coding assistant (Claude, Anthropic) was used to help write and review the analysis code, run and document the experiments, and draft and edit parts of the text. All reported numbers are produced by the released code (Section 10). The authors reviewed all code, analyses and text and take full responsibility for the content.

## 5 Public Provider-Labelled Data

### 5.1 Data

We use the public "Healthcare Provider Fraud Detection Analysis" dataset [47] (CC0 public domain), which its publisher describes as Medicare inpatient and outpatient claims linked to beneficiaries, with each provider labelled as potentially fraudulent or not; graph models studied on these data are discussed in Section 2. Only the labelled training files are used: 558,211 claims (40,474 inpatient, 517,737 outpatient) dated between November 2008 and December 2009, filed by 5,410 providers (506, or 9.4%, labelled fraudulent) for 138,556 beneficiaries, with 100,737 distinct attending, operating or other physicians. The documentation describes neither the source of the claims nor how the labels were obtained, so we treat them as given and discuss this

limitation in Section 9.6. Because labels exist only per provider, the prediction unit is the provider; claim-level ring detection cannot be evaluated on these data.

## 5.2 Features and graph

**P** (23 provider aggregates): claim volume and inpatient share, beneficiaries and claims per beneficiary, distinct attending and operating physicians, reimbursement totals and log-amount statistics, deductibles, claim duration and length of stay, diagnosis and procedure counts, share of claims with an operating physician, and the age, chronic-condition burden, mortality, renal-disease share, annual reimbursements and geographic spread of the provider's beneficiaries. **G** (8 label-free graph features): for both shared beneficiaries and shared physicians, the number of co-providers, the maximum Jaccard overlap with another provider, the share of the provider's beneficiaries (physicians) seen by other providers, and their mean number of providers. **L** (4 label-neighbourhood features): the share of co-providers that are known fraudulent *training* providers, and an indicator of having any labelled co-provider, through beneficiaries and through physicians. L uses only training labels and never a provider's own label.

The heterogeneous graph has provider, beneficiary and physician nodes, with relations *serves* (provider–beneficiary, 363,300 edges) and *employs* (provider–physician) plus reverse relations. Physicians working for a single provider carry no cross-provider information and are summarised in P rather than kept as nodes, leaving 7,685 shared physicians (16,287 edges). Provider nodes carry P (or P+G for the hybrids), beneficiary nodes 22 demographic and clinical attributes, and physician nodes their claim and provider counts. HAN uses the provider-only meta-paths provider–beneficiary–provider and provider–physician–provider.

## 5.3 Protocol

All providers' claims are observed and define the graph; labels of validation and test providers are hidden. This matches an audit in which claims are available for every provider but only some providers have been investigated. Providers are assigned to five stratified folds; in rotation $k$, fold $k$ is the test set, fold $k+1$ the validation set and the remaining three folds the training set, so every provider is tested exactly once. Hyperparameters are selected once, with the same budget of six configurations per method, on a separate stratified 60/20/20 split (seed 1000), and then fixed for the five rotations. Graph models are trained full-batch for at most 100 epochs with early stopping (patience 10) on validation AP. Metrics are AP, ROC-AUC, F1 at the validation-selected threshold, and recall among the top 10% of providers by score (the label prevalence is 9.4%). Because the rotations' training sets overlap, their results are positively correlated and a plain paired $t$-test over them is anti-conservative. Differences between methods are therefore tested with the corrected resampled $t$-test [48], which replaces the factor $1/J$ in the variance of the mean difference by $1/J + n_2/n_1$, where $J = 5$ is the number of rotations and $n_2/n_1 = 1/3$ the ratio of test to training providers; $p$-values are Holm-adjusted over all comparisons. With five rotations these tests have little power, so we also report a provider-level bootstrap of the difference in AP from the best method (1,000 resamples within each rotation's test fold

and label, trained models held fixed); it reflects which providers happen to be tested, not retraining variability. To gain power, we repeated the five-fold rotation four more times with reshuffled folds (25 rotations in all) for the tabular models, the MLP and three GNNs (SAGE, HAN and HGT; RGCN, Simple-HGN and the hybrids were omitted to save compute), and compared each method with GB-P by the same corrected test with $J = 25$ (24 degrees of freedom) [48, 49]. Because the tuning split shares providers with the test folds, we also re-selected every method's configuration inside each rotation of the first repetition, with the same budget and using only that rotation's validation fold (nested selection; the hybrids inherit their base GNN's choice). Both analyses belong to the pre-specified second round of experiments (Section 4.5).

# 6 Results on the Controlled Benchmark

Each test period contains on average 4,943 claims, of which 384 belong to four unseen rings and 133 are isolated fraud, so the chance levels of ring AP and isolated AP are 0.080 and 0.029. Table 1 reports ring AP for every method and regime over the twelve seeds, Fig. 3 shows the main methods with 95% intervals, and Fig. 4 summarises paired differences. Table 2 averages the paired differences over the nine regimes, with the seed as the unit (eleven degrees of freedom). Recall when the top 5% of claims are investigated cannot exceed 0.479, because fraud makes up 10.5% of test claims, and every method with aggregate features or a graph reached 0.43–0.48 in every regime; at this budget the methods differ little (supplementary results), so we compare them by AP, which covers the whole ranking.

**Table 1** Ring-fraud detection on unseen test rings: average precision of ring versus legitimate claims (chance level 0.08), mean over twelve worlds per regime. Bold: best per regime. Standard deviations are in the supplementary results files.

| | $\delta = 0$ | | | $\delta = 1$ | | | $\delta = 2$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Method / $s$ | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 |
| GB-C | 0.082 | 0.083 | 0.084 | 0.327 | 0.327 | 0.324 | 0.807 | 0.809 | 0.807 |
| MLP-C | 0.089 | 0.083 | 0.088 | 0.345 | 0.349 | 0.352 | 0.800 | 0.804 | 0.805 |
| LR-CA | 0.903 | 0.871 | 0.856 | 0.964 | 0.955 | 0.947 | 0.989 | 0.986 | 0.986 |
| RF-CA | 0.877 | 0.851 | 0.847 | 0.947 | 0.928 | 0.937 | 0.984 | 0.982 | 0.984 |
| GB-CA | 0.919 | 0.896 | 0.886 | 0.953 | 0.936 | 0.938 | 0.986 | 0.985 | 0.986 |
| GB-CAG | 0.924 | 0.970 | **0.990** | 0.949 | 0.976 | 0.992 | 0.989 | 0.993 | 0.996 |
| SAGE | 0.943 | 0.957 | 0.971 | 0.974 | 0.981 | 0.990 | 0.993 | 0.992 | 0.992 |
| RGCN | 0.935 | 0.950 | 0.955 | 0.968 | 0.979 | 0.984 | 0.994 | 0.994 | 0.995 |
| HAN | 0.900 | 0.899 | 0.921 | 0.958 | 0.961 | 0.967 | 0.989 | 0.987 | 0.988 |
| Simple-HGN | 0.917 | 0.947 | 0.953 | 0.968 | 0.978 | 0.981 | 0.992 | 0.992 | 0.993 |
| HGT | 0.923 | 0.933 | 0.955 | 0.961 | 0.976 | 0.977 | 0.993 | 0.993 | 0.993 |
| SAGE+AG | **0.962** | **0.979** | 0.988 | **0.974** | **0.986** | 0.995 | 0.994 | **0.996** | **0.997** |
| HGT+AG | 0.959 | 0.976 | 0.990 | 0.973 | 0.984 | **0.996** | **0.994** | 0.995 | 0.997 |
| Overlap score | 0.283 | 0.906 | 0.982 | 0.305 | 0.903 | 0.980 | 0.317 | 0.896 | 0.978 |

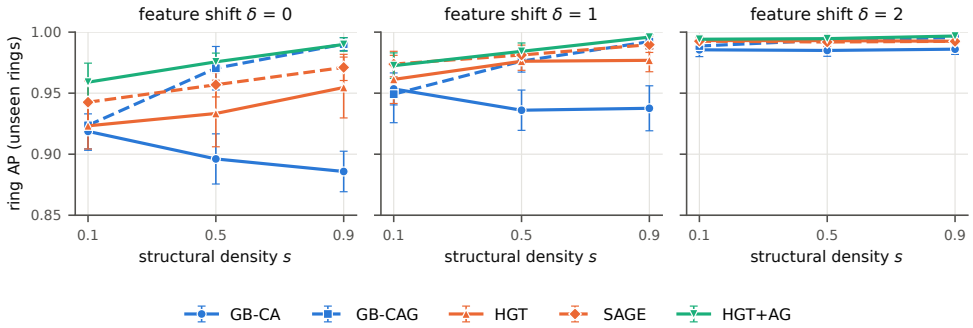



**Fig. 3** Ring AP on unseen test rings by feature shift $\delta$ and structural density $s$ (mean and 95% $t$-interval over twelve worlds). GB-CA: gradient boosting on claim and aggregate features; GB-CAG: plus engineered graph features; HGT+AG: HGT whose claim nodes also receive the A and G features. Feature-only models (GB-C, MLP-C) lie far below the plotted range (Table 1).

## 6.1 Claim features alone do not reveal rings

When ring claims are drawn like legitimate claims ($\delta = 0$), models that see only the claim's own attributes perform at chance (GB-C 0.082–0.084, MLP-C 0.083–0.089). As $\delta$ grows their ring AP rises to about 0.33 ($\delta = 1$) and 0.81 ($\delta = 2$). This confirms that the benchmark separates the two sources of evidence as intended.

## 6.2 At default ring volume, entity aggregates capture most ring evidence

Adding per-period aggregates of the claim's patient, doctor and hospital (GB-CA) raises ring AP at $\delta = 0$ to 0.886–0.919. This is a consequence of the generator's default ring volume rather than a general property of collusion: ring patients file almost six times as many claims as other patients and see more doctors, and ring doctors file bursts of claims (Section 3.1). Simple counts capture these signals in every regime; when ring volume is reduced to four claims per month, aggregates detect almost nothing (Section 6.6). The remaining gap at default volume is where coordination structure matters.

## 6.3 Coordination structure helps only when the ring block is dense

Engineered graph features add little at low density but substantially at moderate and high density: at $\delta = 0$, GB-CAG improves on GB-CA by +0.005 ($s = 0.1$, n.s.), +0.074 ($s = 0.5$) and +0.104 ($s = 0.9$; both Holm $p < 0.001$). The unsupervised overlap score tells the same story: it is weak at $s = 0.1$ (0.283) but reaches 0.906 and 0.982 at $s = 0.5$ and 0.9 without any training. GNNs show the same pattern: SAGE improves on GB-CA by +0.085 at $\delta = 0$, $s = 0.9$ (Holm $p < 0.001$). Averaged over the nine regimes, every graph-aware method outperforms GB-CA (GB-CAG +0.033, SAGE +0.034, RGCN +0.030, Simple-HGN +0.026, HGT +0.024, HAN +0.010, hybrids +0.043 and +0.042; Holm $p \leq 0.020$ with the seed as unit). When ring claims are

strongly anomalous ($\delta = 2$), all methods that see claim and aggregate features exceed 0.98 and the differences become negligible.

## 6.4 GNNs do not outperform gradient boosting with graph features

The central comparison is between message passing and the same structural information supplied as engineered features. Averaged over the regimes, heterogeneous GraphSAGE matches GB-CAG ($\Delta = +0.001$, 95% CI $[-0.003, 0.006]$), RGCN, Simple-HGN and HGT have small negative point estimates ($-0.003$, $-0.006$ and $-0.008$; Holm $p \geq 0.18$), and HAN is significantly worse ($-0.023$, Holm $p = 0.002$; Table 2). Every upper CI limit is at most $+0.006$, so the data exclude a pure-GNN advantage larger than that. Within single regimes, a pure GNN was significantly better than GB-CAG once: RGCN at $\delta = 2$, $s = 0.1$ ($+0.005$, Holm $p = 0.007$), where every method exceeded 0.98. Pure GNNs were significantly worse in eight method–regime pairs, mostly for dense rings and for HAN (Holm-adjusted within each nine-regime family).

The hybrids, which add message passing on top of the engineered features, gained a small but statistically reliable amount over GB-CAG: $+0.010$ $[0.005, 0.015]$ for SAGE+AG (Holm $p = 0.006$) and $+0.009$ $[0.004, 0.015]$ for HGT+AG ($p = 0.016$). The gain was largest for sparse rings whose claims look legitimate ($\delta = 0$, $s = 0.1$: $+0.038$ and $+0.035$, Holm $p = 0.007$ and 0.018), where the engineered features add least, and a pre-declared test on fresh seeds confirmed it at low ring volume (Section 6.6). Message passing therefore carries some information beyond our engineered features, most of it for sparse rings, but it does not replace them.

**Table 2** Paired difference in average precision versus GB-CAG, averaged over the nine regimes: mean $\Delta \pm$ half-width of the 95% $t$-interval over the twelve generator seeds, and Holm-adjusted paired $t$-test $p$ (within each metric). The nine regimes of a seed share its legitimate claims, isolated fraud and ring members, so the seed, not the world, is the unit of replication.

| | Ring AP | | Isolated AP | | Overall AP | |
|---|---|---|---|---|---|---|
| Method | $\Delta \pm$ CI | $p$ | $\Delta \pm$ CI | $p$ | $\Delta \pm$ CI | $p$ |
| SAGE | $+0.001 \pm 0.005$ | 0.527 | $-0.097 \pm 0.023$ | <0.001 | $-0.008 \pm 0.004$ | 0.004 |
| RGCN | $-0.003 \pm 0.004$ | 0.334 | $-0.109 \pm 0.028$ | <0.001 | $-0.012 \pm 0.004$ | <0.001 |
| HAN | $-0.023 \pm 0.010$ | 0.002 | $-0.138 \pm 0.015$ | <0.001 | $-0.030 \pm 0.007$ | <0.001 |
| Simple-HGN | $-0.006 \pm 0.007$ | 0.188 | $-0.093 \pm 0.021$ | <0.001 | $-0.014 \pm 0.004$ | <0.001 |
| HGT | $-0.008 \pm 0.008$ | 0.188 | $-0.124 \pm 0.018$ | <0.001 | $-0.017 \pm 0.006$ | <0.001 |
| SAGE+AG | $+0.010 \pm 0.005$ | 0.006 | $-0.087 \pm 0.022$ | <0.001 | $0.000 \pm 0.004$ | 0.985 |
| HGT+AG | $+0.009 \pm 0.005$ | 0.016 | $-0.113 \pm 0.022$ | <0.001 | $-0.003 \pm 0.004$ | 0.374 |
| MLP-C | $-0.563 \pm 0.018$ | <0.001 | $-0.025 \pm 0.016$ | 0.006 | $-0.407 \pm 0.022$ | <0.001 |

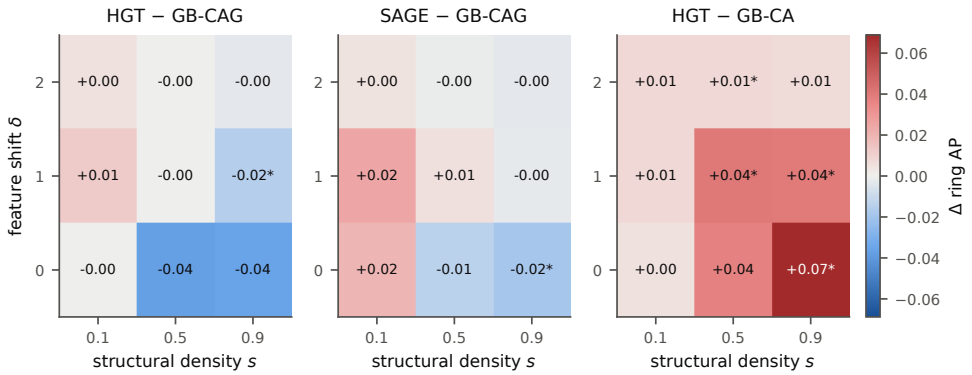



**Fig. 4** Mean paired difference in ring AP (twelve worlds per cell). Left and middle: HGT and SAGE versus gradient boosting with engineered graph features. Right: HGT versus gradient boosting without graph features. *Holm-adjusted $p < 0.05$ within the nine-regime family.

## 6.5 Message passing dilutes claim-level anomalies

Isolated fraud shows the opposite pattern (Table 3). Every GNN is significantly worse than GB-CAG, by 0.093 (Simple-HGN) to 0.138 (HAN) AP averaged over regimes (Holm $p < 0.001$ with the seed as unit), with HAN and HGT the weakest. The MLP control, which sees the claim's own attributes and the static attributes of its patient, doctor and hospital—the GNNs' inputs without degree encodings or message passing—comes within 0.025 of GB-CAG ($p = 0.006$). Aggregating a claim's representation with those of its neighbours, which are overwhelmingly legitimate, blurs the claim's own anomaly; the hybrids, whose claim nodes also see the engineered ratios, recover little of the gap (−0.087 and −0.113). On all fraud combined (overall AP), GB-CAG has higher AP than every pure GNN, by 0.008–0.030 (all Holm $p \leq 0.004$), and does not differ from the hybrids ($p \geq 0.37$).

## 6.6 When ring members are not unusually active

At default volume, ring members stand out by how many claims they file in every regime (Section 3.1). To test which conclusions depend on this, we repeated the three $\delta = 0$ regimes with ring volume reduced from 16 to 4 claims per month and everything else unchanged. In a test period a ring patient then files 3.7 claims, about the 95th percentile of other patients, a ring doctor 20 claims against 16, and rings make up 2.1% of the non-isolated claims, which is also the chance level of ring AP (Table 4). All 14 methods were run with the hyperparameters of the main experiment and, to check that these did not handicap any method, again after repeating the tuning protocol on low-volume development worlds.

With the hyperparameters of the main experiment, three results change. First, per-entity aggregates no longer detect rings: GB-CA reached a ring AP of 0.049–0.055, barely above chance. Second, coordination structure becomes the main source of evidence for moderately and densely linked rings: GB-CAG reached 0.409 and 0.570 at $s = 0.5$ and 0.9, 0.36 and 0.51 above GB-CA (Holm $p \leq 0.001$), against 0.066 at

**Table 3** Isolated-fraud detection: average precision of isolated-fraud versus legitimate claims (chance level 0.03), mean over twelve worlds per regime. Bold: best per regime.

| | $\delta = 0$ | | | $\delta = 1$ | | | $\delta = 2$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Method / $s$ | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 |
| GB-C | 0.483 | 0.494 | 0.482 | 0.574 | 0.566 | 0.556 | 0.651 | 0.644 | 0.640 |
| MLP-C | 0.529 | 0.529 | 0.542 | **0.611** | 0.607 | 0.608 | 0.648 | 0.656 | 0.652 |
| LR-CA | 0.342 | 0.299 | 0.282 | 0.553 | 0.538 | 0.516 | **0.668** | 0.657 | **0.655** |
| RF-CA | 0.553 | 0.522 | 0.492 | 0.592 | 0.566 | 0.558 | 0.629 | 0.611 | 0.607 |
| GB-CA | 0.508 | 0.518 | 0.515 | 0.577 | 0.560 | 0.564 | 0.651 | 0.619 | 0.625 |
| GB-CAG | **0.559** | **0.604** | **0.627** | 0.609 | **0.609** | **0.626** | 0.664 | **0.660** | 0.651 |
| SAGE | 0.414 | 0.418 | 0.493 | 0.524 | 0.528 | 0.547 | 0.611 | 0.603 | 0.602 |
| RGCN | 0.408 | 0.440 | 0.472 | 0.503 | 0.511 | 0.504 | 0.603 | 0.597 | 0.594 |
| HAN | 0.390 | 0.406 | 0.411 | 0.476 | 0.477 | 0.477 | 0.584 | 0.573 | 0.572 |
| Simple-HGN | 0.403 | 0.412 | 0.497 | 0.548 | 0.516 | 0.525 | 0.634 | 0.625 | 0.615 |
| HGT | 0.339 | 0.387 | 0.449 | 0.509 | 0.486 | 0.503 | 0.618 | 0.611 | 0.594 |
| SAGE+AG | 0.423 | 0.480 | 0.569 | 0.477 | 0.523 | 0.577 | 0.586 | 0.591 | 0.605 |
| HGT+AG | 0.352 | 0.447 | 0.566 | 0.421 | 0.499 | 0.557 | 0.557 | 0.587 | 0.608 |
| Overlap score | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 |

$s = 0.1$, where no method was far above chance. Third, the pure GNNs recovered little of this structure: at $s \geq 0.5$ their ring AP was 0.058–0.151, only 0.01–0.07 above GB-CA averaged over densities, and 0.23–0.29 below GB-CAG (Holm $p \leq 0.002$ with the seed as unit). One result does not change: no method had significantly higher ring AP than GB-CAG at any density, and the hybrids, which receive the engineered features, matched it (SAGE+AG +0.001, 95% CI $[-0.059, 0.061]$; HGT+AG −0.041 $[-0.098, 0.015]$). On isolated fraud the GNNs were again below GB-CAG, by 0.03–0.08, though not significantly with six seeds.

Re-tuning on low-volume development worlds changed the selected configuration of most methods (HAN, RGCN and Simple-HGN kept theirs and reproduced their results exactly) but not these conclusions (Table 4, right). GB-CAG improved slightly (0.445 and 0.600 at $s = 0.5$ and 0.9, 0.39 and 0.55 above GB-CA), the pure GNNs stayed between 0.058 and 0.162 at $s \geq 0.5$ and 0.26–0.31 below GB-CAG averaged over densities (Holm $p \leq 0.001$), and SAGE+AG matched GB-CAG on average (+0.009 $[-0.048, 0.065]$). The only significant advantage over GB-CAG in either setting was that of SAGE+AG for sparse rings after re-tuning ($s = 0.1$: 0.129 against 0.078, Holm $p = 0.035$), where every method was close to chance. It echoes the default-volume result, where the hybrids' largest gain was also for sparse rings, which suggests that message passing on top of the engineered features helps most where those features are weakest.

Because this advantage was found by exploring seeds 1–6, we tested it once more on twelve fresh seeds (7–18), with the comparison, test and decision rule fixed beforehand (Online Resource 1, Section S4). SAGE+AG again exceeded GB-CAG, by +0.024 $[0.005, 0.043]$ ($p = 0.016$; nine of twelve seeds), about half the exploratory estimate. The advantage is real but small, and both methods stayed close to chance (0.129 and 0.104 against 0.021).

**Table 4** Ring AP on unseen test rings at $\delta = 0$ when ring volume is reduced from 16 to 4 claims per month (mean over six worlds per cell). At low volume a ring patient files about as many claims as the 95th percentile of other patients and a ring doctor 1.3 times as many as other doctors (default: 5.7 and 2.5 times the averages), and the chance level of ring AP falls from 0.080 to 0.021. Frozen: hyperparameters tuned at default volume; re-tuned: the tuning protocol repeated on low-volume development worlds (hybrids inherit their base configuration). Bold: best per column.

| | Default volume | | | Low volume, frozen | | | Low volume, re-tuned | | |
|---|---|---|---|---|---|---|---|---|---|
| Method / $s$ | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 |
| GB-C | 0.084 | 0.086 | 0.087 | 0.024 | 0.023 | 0.022 | 0.023 | 0.023 | 0.022 |
| MLP-C | 0.088 | 0.087 | 0.095 | 0.024 | 0.024 | 0.024 | 0.024 | 0.023 | 0.024 |
| LR-CA | 0.923 | 0.888 | 0.868 | **0.106** | 0.087 | 0.088 | 0.090 | 0.083 | 0.089 |
| RF-CA | 0.875 | 0.838 | 0.844 | 0.057 | 0.064 | 0.064 | 0.062 | 0.068 | 0.073 |
| GB-CA | 0.925 | 0.905 | 0.889 | 0.049 | 0.052 | 0.055 | 0.047 | 0.052 | 0.051 |
| GB-CAG | 0.930 | 0.983 | **0.990** | 0.066 | **0.409** | **0.570** | 0.078 | **0.445** | **0.600** |
| SAGE | 0.938 | 0.963 | 0.976 | 0.074 | 0.108 | 0.131 | 0.065 | 0.112 | 0.162 |
| RGCN | 0.933 | 0.953 | 0.954 | 0.066 | 0.112 | 0.142 | 0.066 | 0.112 | 0.142 |
| HAN | 0.908 | 0.906 | 0.921 | 0.062 | 0.065 | 0.058 | 0.062 | 0.065 | 0.058 |
| Simple-HGN | 0.915 | 0.960 | 0.957 | 0.074 | 0.133 | 0.151 | 0.074 | 0.133 | 0.151 |
| HGT | 0.910 | 0.948 | 0.949 | 0.046 | 0.079 | 0.098 | 0.050 | 0.082 | 0.108 |
| SAGE+AG | **0.973** | **0.985** | 0.983 | 0.103 | 0.379 | 0.566 | **0.129** | 0.420 | 0.600 |
| HGT+AG | 0.964 | 0.984 | 0.990 | 0.074 | 0.324 | 0.523 | 0.074 | 0.301 | 0.477 |
| Overlap score | 0.254 | 0.937 | 0.985 | 0.057 | 0.194 | 0.281 | 0.057 | 0.194 | 0.281 |

Figure 5 traces the transition over four ring volumes (16, 8, 4 and 2 claims per month; seeds 1–6, hyperparameters of the main experiment; full results in Online Resource 1, Table S3). The value of coordination structure is not monotonic in volume. At 16 claims per month aggregates already find most rings and graph features add 0.06 AP averaged over densities. At 8, where ring patients file about as many claims as the 99th percentile of other patients, aggregates fall to 0.31–0.42 and graph features add 0.42 (0.51 and 0.61 for moderately and densely linked rings); at 4 they add 0.30; and at 2 claims per month, one claim per ring member, no method is far above chance. Pure GNNs fall further behind GB-CAG as volume decreases, by 0.01–0.06 AP at 16, 0.10–0.29 at 8 and 0.23–0.29 at 4 claims per month, whereas the hybrids stay within 0.05 of it at every volume.

## 6.7 Routing claims to a GNN adds little beyond combining scores

In the design usually proposed, GB-CAG filters and the GNN decides (re-rank). The routed system matched GB-CAG (GB-CAG→HGT $\Delta$AP $= -0.001$, 95% CI $[-0.006, 0.003]$; GB-CAG→SAGE $+0.001$), and escalating more claims to HGT lowered both ring and isolated AP, because HGT ranks both less well than GB-CAG. Ordering escalated claims by the mean of the two models' percentiles did better: GB-CAG→SAGE gained $+0.006$ $[0.003, 0.010]$ (Holm $p = 0.042$, positive for all six seeds), but validation then chose to fuse *every* claim in 30 of the 54 worlds, so the gain comes from combining two rankings, not from routing a subset of claims. At the ring level

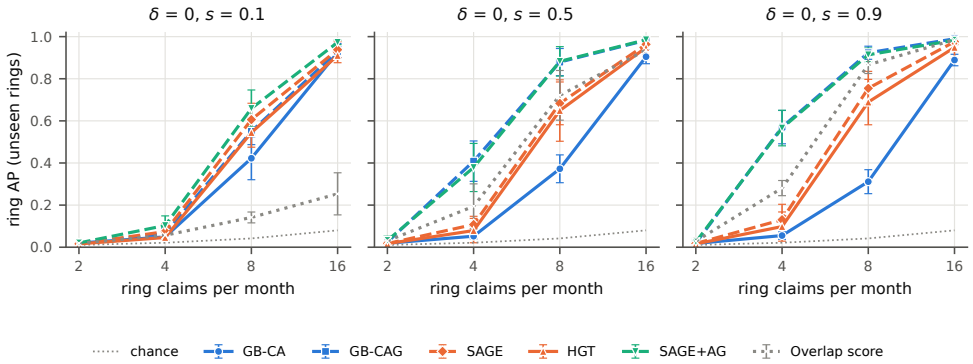



**Fig. 5** Ring AP on unseen test rings at $\delta = 0$ against ring volume (claims per ring per month; 16 is the default), mean and 95% $t$-interval over six worlds (seeds 1–6), hyperparameters of the main experiment. Dotted line: chance level. Structure adds most at intermediate volume; at two claims per month every method is close to chance. GB-CAG is largely hidden behind SAGE+AG, with which it nearly coincides.

all methods reached the ceiling: every test ring had at least one claim among the top 5% of claims (Online Resource 1, Section S2, Table S2 and Fig. S1).

### 6.8 Cost

Median training time per world was 3 s for GB-CAG and 39–144 s for the GNNs, 13–50 times longer (laptop CPU with several jobs in parallel, so the times are indicative), and scoring a whole audit period took at most 0.25 s for any method. On an idle single thread, scoring one claim took a median of 2.1 ms with GB-CAG on a precomputed feature row and 10.7 ms with HGT on the claim's exact three-hop computation subgraph. These figures exclude keeping aggregates, graph features and degree counts up to date as claims arrive (details in Online Resource 1, Section S5).

## 7 Mechanism and Memorisation

### 7.1 What isolated and ring fraud look like

Fig. 6 shows the test period of one world (seed 1, $\delta = 0$) with the outcomes of GB-CAG and HGT at their validation-selected thresholds. An isolated fraud claim sits on an ordinary patient–doctor edge, so nothing in the graph distinguishes it and only its own attributes can. A ring at $s = 0.1$ links each recruited patient to one or two of the four ring doctors, a pattern that resembles ordinary shared care, whereas at $s = 0.9$ the same members form an almost complete block. The two worlds share the recruited ring members and every legitimate and isolated-fraud claim, and differ only in which ring links exist. In this world GB-CAG flagged 81 and 91 of the ring's 96 claims at $s = 0.1$ and $s = 0.9$, and HGT 62 and 94: the graph model gains most where the block is dense, as in the averages of Table 1. The figure illustrates one world only; the quantitative statements rest on all 54.

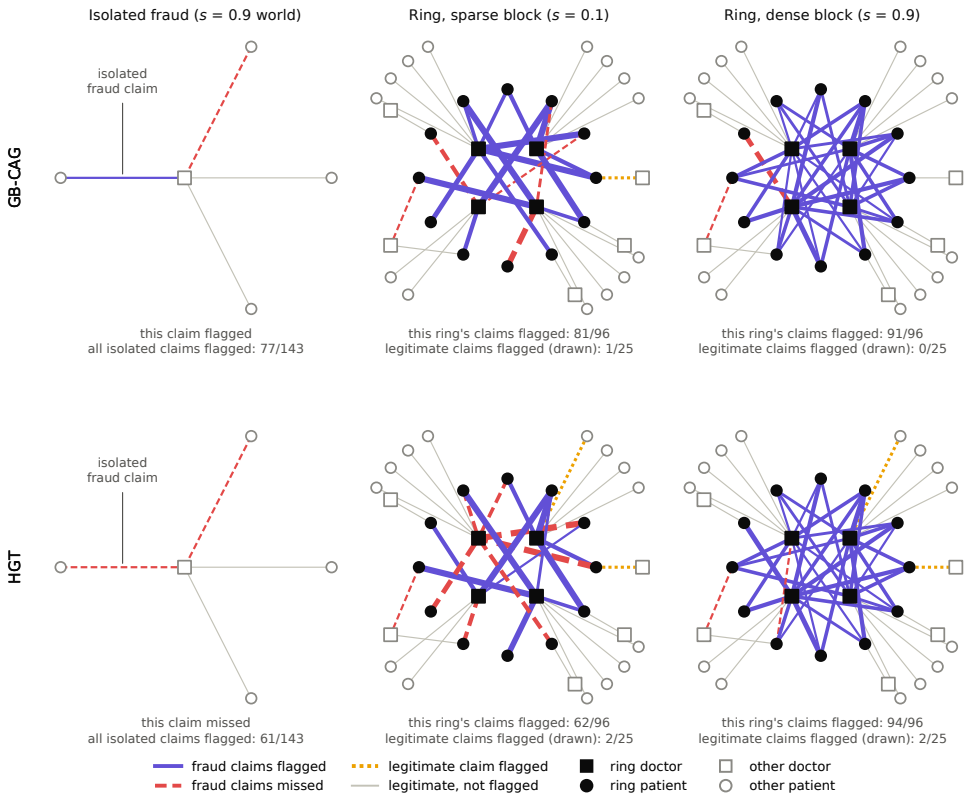



**Fig. 6** Isolated and ring fraud in one benchmark world (seed 1, $\delta = 0$; test period), with model outcomes at each model's validation-selected threshold (top: GB-CAG; bottom: HGT). Nodes carry the ground truth (filled: ring member; squares: doctors; circles: patients); edges are patient–doctor pairs with claims in the period, coloured by the outcome for the majority of their claims (line width: number of claims). Left: an isolated fraud claim drawn at random, its patient's doctors and up to ten other patients of its doctor. Middle and right: one test ring, every doctor its patients visit and up to four other patients per ring doctor, at low and high structural density; the two worlds share all recruited members and all legitimate claims. The figure illustrates the generator's typologies; it is not evidence that real rings look alike.

## 7.2 What the models need

Table 5 reports ablations in the regime where structure matters most ($\delta = 0$, $s = 0.9$), at the default ring volume (with 95% intervals in Online Resource 1, Fig. S2). Degree-preserving rewiring—permuting which doctor bills each claim within a period, which keeps every patient's and doctor's claim volume but destroys the ring's co-occurrence block—reduces every graph-aware method to the same level (0.757–0.770), close to gradient boosting without graph features after rewiring (0.748). The GNNs therefore do exploit coordination structure, and once it is removed nothing distinguishes them from a model that sees volume alone.

Depth matters: with a single message-passing layer, ring AP falls by 0.135 (SAGE) and 0.161 (HGT); two and three layers perform alike. Removing the direct patient–doctor relation costs little (−0.016 and −0.047, not significant after correction), because two-hop paths through claim nodes carry the same information. The largest effect is specific to HGT: without degree encodings its ring AP drops from 0.949 to 0.569 (−0.379, Holm $p = 0.003$), whereas sum-aggregating SAGE changes little (−0.029, n.s.). Softmax attention normalises over neighbours and so cannot count them; supplying counts as input features, or aggregating by summation, restores the signal [38].

**Table 5** Mechanism ablations at $\delta = 0$, $s = 0.9$ (six worlds): ring AP (mean ± s.d.) and paired change from the full model, with Holm-adjusted $p$ over every ablation comparison in this table and Table S4 of Online Resource 1, which lists HGT's single-factor variants. The tuned GNNs use three layers.

| Method | Variant | Ring AP | Δ vs. full |
|---|---|---|---|
| GB-CA | Full model | 0.889 ± 0.026 | – |
| | Degree-preserving rewiring | 0.748 ± 0.027 | −0.141 ($p = 0.018$) |
| | + ring identifier (legacy leak) | 1.000 ± 0.000 | +0.111 ($p = 0.003$) |
| GB-CAG | Full model | 0.990 ± 0.008 | – |
| | Degree-preserving rewiring | 0.757 ± 0.046 | −0.232 ($p = 0.002$) |
| SAGE | Full model | 0.976 ± 0.013 | – |
| | Degree-preserving rewiring | 0.759 ± 0.034 | −0.217 ($p = 0.002$) |
| | No patient–doctor relation | 0.960 ± 0.019 | −0.016 ($p = 0.360$) |
| | No degree encodings | 0.947 ± 0.037 | −0.029 ($p = 1.000$) |
| | 1 message-passing layer | 0.841 ± 0.034 | −0.135 ($p = 0.002$) |
| | 2 layers | 0.957 ± 0.023 | −0.019 ($p = 0.413$) |
| HGT | Full model | 0.949 ± 0.052 | – |
| | Degree-preserving rewiring | 0.770 ± 0.029 | −0.179 ($p = 0.036$) |
| | No reverse relations (legacy graph) | 0.772 ± 0.024 | −0.177 ($p = 0.022$) |

Single-factor ablations of HGT (Online Resource 1, Section S6 and Table S4) point the same way. Depth beyond two layers and the number of attention heads made no significant difference, so there is no sign of over-smoothing up to four layers. With claim attributes set to zero, ring AP was unchanged and isolated AP fell to chance; an HGT that saw only the degree encodings and the graph reached a ring AP of $0.982 \pm 0.018$, as high as the full model, and detected almost no isolated fraud. HGT's ring detection is therefore structural—counts of who bills whom, propagated along patient–claim–doctor paths—which is the information that the engineered shared-patient features summarise directly. At default volume these counts also carry the ring members' excess claims; Section 6.6 shows that when they do not, the GNNs fall far behind gradient boosting with the engineered features. Permuting groups of inputs of the trained models (Online Resource 1, Section S7) is consistent with this: GB-CAG relied on aggregates when rings were sparse and on the graph features, above all the normalised triadic closure of the claim, when they were dense, whereas SAGE and HGT relied almost entirely on degree encodings.

## 7.3 Two evaluation pitfalls, quantified

The mechanism ablations (Table 5) also quantify two defects that are easy to introduce. Adding the generator's ring identifier to the feature set—a column that is non-zero only for ring claims—raises gradient boosting from 0.889 to a perfect 1.000. Building the heterogeneous graph with one-directional relations only (patient→claim→doctor→hospital) leaves claim nodes reachable only from patients, so doctor context never reaches them; HGT then loses 0.177 AP. Both defects produce plausible-looking numbers and neither is caught by aggregate metrics alone; the tests listed in Section 10 detect both.

## 7.4 Recognising known members is not detecting rings

Table 6 (and Online Resource 1, Fig. S3) compares new rings (emergent protocol) with rings whose members recur from training (persistent protocol). The MLP control, which sees only the claim and the static attributes of its patient, doctor and hospital, rises from chance to 0.692 ($s = 0.1$) and 0.591 ($s = 0.9$) when rings recur: it identifies the members from their attributes. HGT gains 0.056 at $s = 0.1$ (95% CI $[0.029, 0.084]$), and gradient boosting with aggregates gains 0.035–0.063, while the unsupervised overlap score, which cannot memorise, is unchanged. Evaluations in which test rings share members with training rings therefore overstate how well a model detects coordination, by an amount that differs between methods and can reorder them.

**Table 6** Ring AP when test rings are new (emergent) versus recurring members of training rings (persistent), $\delta = 0$, six worlds each; gain = persistent − emergent (paired over worlds, 95% $t$-interval). The overlap score is unsupervised and cannot memorise.

| | $s = 0.1$ | | | $s = 0.9$ | | |
|---|---|---|---|---|---|---|
| Method | new | recurring | gain | new | recurring | gain |
| MLP-C | 0.088 | 0.692 | +0.604 [+0.523, +0.685] | 0.095 | 0.591 | +0.496 [+0.416, +0.577] |
| GB-CA | 0.925 | 0.959 | +0.035 [+0.014, +0.055] | 0.889 | 0.952 | +0.063 [+0.034, +0.092] |
| GB-CAG | 0.930 | 0.955 | +0.025 [−0.012, +0.062] | 0.990 | 0.994 | +0.005 [−0.002, +0.012] |
| SAGE | 0.938 | 0.969 | +0.031 [−0.019, +0.081] | 0.976 | 0.978 | +0.002 [−0.015, +0.020] |
| HGT | 0.910 | 0.967 | +0.056 [+0.029, +0.084] | 0.949 | 0.981 | +0.032 [−0.017, +0.082] |
| Overlap score | 0.254 | 0.256 | +0.002 [−0.137, +0.142] | 0.985 | 0.991 | +0.006 [−0.009, +0.020] |

# 8 Results on the Public Provider-Labelled Data

Table 7 and Fig. 7 report provider-level results over the five rotations (1,082 test providers each, about 101 fraudulent). Gradient boosting on provider aggregates (GB-P) has the highest mean AP ($0.704 \pm 0.045$). Adding label-free graph features (GB-PG, $-0.009$, corrected 95% CI $[-0.042, 0.023]$) or graph and label-neighbourhood features (GB-PGL, $-0.007$ $[-0.030, 0.016]$) does not improve it. Every GNN has lower AP than GB-P, by 0.029 (HGT) to 0.083 (RGCN), and the hybrids, which give the GNN the provider and graph features, are no better than the plain GNNs. With five overlapping rotations, none of the 24 differences is significant after the correction and Holm adjustment (smallest Holm $p = 0.13$, SAGE+G versus GB-P); a plain paired $t$-test, which ignores the overlap, would have called SAGE+G significantly worse (Holm $p = 0.021$). Holding the trained models fixed, the provider-level bootstrap places the AP of every GNN and hybrid, and of GB-PG, below that of GB-P (95% intervals from $[-0.108, -0.055]$ for RGCN to $[-0.048, -0.006]$ for HGT), so the shortfall does not depend on which providers happened to be tested. ROC-AUC lies between 0.927 and 0.942 for all methods and recall among the top 10% of providers between 0.59 and 0.64, so the methods rank providers similarly; the AP differences come from the top of the ranking. Under the same parallel load as before, median training time per rotation was 2–15 minutes for the GNNs (range 1.4–24 minutes) against 7 s for GB-P.

The two pre-specified checks of this protocol left the picture unchanged (Online Resource 1, Section S9). Over 25 rotations (five repetitions of the five-fold rotation), GB-P again had the highest mean AP (0.693), graph features changed it by $-0.007$ $[-0.019, 0.006]$ (GB-PG) and $-0.006$ $[-0.019, 0.007]$ (GB-PGL), and SAGE, HAN and HGT were 0.025–0.034 below it. No difference was significant after the correction and Holm adjustment (smallest Holm $p = 0.29$, HAN), whereas a plain paired $t$-test over the 25 rotations would have called all eight significant. Re-selecting every method's configuration inside each rotation changed AP by at most 0.010 for the tabular models and the MLP and by $-0.008$ to $+0.036$ for the GNNs and hybrids, none significantly (SAGE chose its shared configuration in every rotation). Contrary to our expectation, the shared selection therefore did not systematically favour the GNNs. With every method selected inside each rotation, GB-P (0.706) remained the best, and every GNN and hybrid was 0.023–0.053 below it (smallest Holm $p = 0.051$).

The graph statistics explain why. Almost every provider (99.5%) shares beneficiaries with other providers (median 30 co-providers), but labels are not assortative through this relation: on average 23.9% of a fraudulent provider's co-providers are labelled fraudulent, against 25.3% for a legitimate provider (both well above the 9.4% base rate because shared beneficiaries concentrate on large providers). Shared physicians show mild assortativity (19.1% versus 14.2%) but connect only 73% of providers, with a median of two co-providers. Fraudulent providers are, however, much larger (median 156 versus 27 claims), a volume signal that provider aggregates capture directly. In the terms of our benchmark, these data resemble the low-density regime, where structure added nothing beyond per-entity aggregates. The labels mark potentially fraudulent providers without identifying rings, so the absence of a graph benefit here does not show that rings are absent from Medicare data; it shows that this public labelling contains no exploitable relational signal for the models tested.

**Table 7** Provider-level fraud detection on the public provider-labelled data (5-fold rotation; mean ± s.d. over folds). F1 at the validation-selected threshold; Recall@10%: share of all fraudulent providers that rank among the top 10% of providers by score.

| Method | AP | ROC-AUC | F1 | Recall@10% |
|---|---|---|---|---|
| LR-P | 0.678 ± 0.041 | 0.940 ± 0.017 | 0.617 ± 0.030 | 0.644 ± 0.023 |
| RF-P | 0.681 ± 0.049 | 0.942 ± 0.017 | 0.624 ± 0.031 | 0.644 ± 0.025 |
| **GB-P** | 0.704 ± 0.045 | 0.939 ± 0.020 | 0.624 ± 0.028 | 0.636 ± 0.027 |
| GB-PG | 0.695 ± 0.040 | 0.937 ± 0.019 | 0.600 ± 0.043 | 0.642 ± 0.024 |
| GB-PGL | 0.697 ± 0.042 | 0.937 ± 0.019 | 0.595 ± 0.045 | 0.634 ± 0.019 |
| MLP-P | 0.685 ± 0.052 | 0.940 ± 0.019 | 0.613 ± 0.028 | 0.642 ± 0.032 |
| SAGE | 0.664 ± 0.047 | 0.938 ± 0.018 | 0.605 ± 0.036 | 0.624 ± 0.046 |
| RGCN | 0.621 ± 0.054 | 0.927 ± 0.019 | 0.578 ± 0.029 | 0.593 ± 0.029 |
| HAN | 0.666 ± 0.046 | 0.938 ± 0.018 | 0.583 ± 0.028 | 0.636 ± 0.045 |
| Simple-HGN | 0.670 ± 0.057 | 0.939 ± 0.020 | 0.600 ± 0.037 | 0.630 ± 0.036 |
| HGT | 0.676 ± 0.058 | 0.939 ± 0.019 | 0.613 ± 0.051 | 0.640 ± 0.051 |
| SAGE+G | 0.653 ± 0.048 | 0.937 ± 0.018 | 0.612 ± 0.040 | 0.623 ± 0.031 |
| HGT+G | 0.648 ± 0.032 | 0.933 ± 0.014 | 0.589 ± 0.022 | 0.615 ± 0.035 |

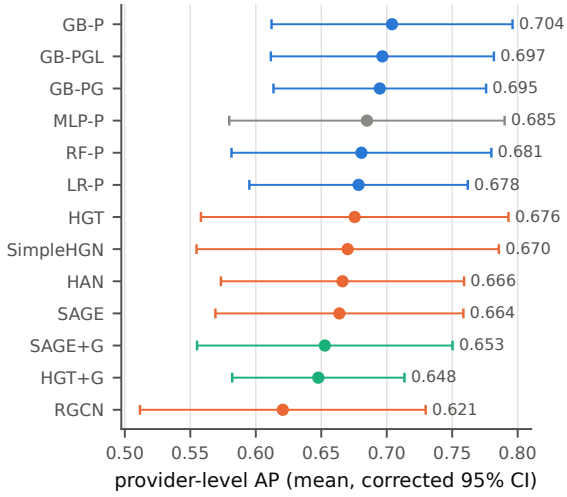



**Fig. 7** Provider-level AP on the public provider-labelled data (mean over five rotations; 95% CI from the corrected resampled $t$-distribution, which accounts for the overlap of the rotations' training sets). Blue: tabular; orange: GNN; aqua: hybrid; grey: MLP.

# 9 Discussion

## 9.1 When does relational structure help?

Table 8 summarises the conditions that emerge from the controlled study. First, the claims must not already be anomalous on their own: when ring claims were strongly upcoded ($\delta = 2$), every model with claim and aggregate features exceeded 0.98 AP and structure became redundant. Second, the value of structure depends on what else

distinguishes ring members: at default volume their elevated utilisation, which per-entity counts capture, already identified most rings, whereas at four to eight claims per month aggregates lost most of their power and coordination structure carried most of the evidence. Third, the trace of coordination must be dense enough to distinguish a ring from ordinary shared care: graph information added 0.07–0.10 AP for moderately and densely linked rings at default volume and 0.36–0.61 at reduced volume, but less for sparse rings (+0.005 at default volume, +0.13 at eight and at most +0.03 at four claims per month). Finally, a ring that files about one claim per member leaves no trace that any of our methods could find.

**Table 8** When relational structure helps: a summary of the controlled results ($\delta = 0$ unless stated). Structure gain: GB-CAG minus GB-CA in ring AP. Volume: ring claims per month (default 16).

| Setting | Main evidence for rings | Structure gain | What works |
|---|---|---|---|
| Ring claims anomalous ($\delta = 2$) | Claim attributes | None: all methods with aggregates $> 0.98$ | Any tabular model with aggregates |
| Members unusually active (volume 16), sparse rings ($s = 0.1$) | Member volume, repeated doctor–patient pairs | +0.005 | GB-CA(G); hybrids add +0.035 to +0.038 |
| Members unusually active (volume 16), dense rings ($s \geq 0.5$) | Volume and shared patients | +0.07 to +0.10 | GB-CAG or hybrids; pure GNNs no better |
| Members moderately active (volume 8), sparse rings | Volume and shared patients | +0.13 | GB-CAG, hybrids and pure GNNs other than HAN alike (0.54–0.66) |
| Members moderately active or not unusual (volume 4–8), dense rings | Shared patients | +0.36 to +0.61 | GB-CAG or hybrids; pure GNNs 0.17–0.54 lower |
| Members not unusual (volume 4), sparse rings | Little | At most +0.03; all methods near chance | Hybrid slightly better (+0.024, confirmed) |
| One claim per member (volume 2) | None detectable | None | No method |
| Public provider-labelled data | Provider volume; no label assortativity | None (GB-PG −0.007, 25 rotations) | Gradient boosting on provider aggregates |

## 9.2 Why GNNs did not beat engineered graph features

When structure did help, gradient boosting with a handful of shared-patient statistics captured it as well as or better than the best GNN. The mechanism analyses suggest why. The useful signal here is largely *counting*—how many of a patient's doctors also share other patients—and attention- or mean-based aggregators cannot count: HGT lost most of its ring detection without degree encodings, whereas sum-aggregating

GraphSAGE barely changed, and an HGT given only degree encodings and the graph detected rings as well as the full model (Section 7). This mirrors the expressiveness argument for sum aggregation [38]. GNNs also dilute claim-level anomalies through neighbourhood averaging, which made them markedly worse on isolated fraud, and they readily memorise the attributes of training-ring members, which inflates performance when rings recur but does not transfer to new rings. Hybrids that combine message passing with the engineered features gained a small but reliable amount over gradient boosting on ring fraud (+0.009 to +0.010 AP), most for sparse rings, where the engineered features are weakest; that is the structure message passing extracts beyond our features, and it comes at 9–40 times the training cost.

The ring-volume experiments sharpen this picture. Once ring members no longer stood out by volume, counting no longer sufficed, and the pure GNNs recovered only a small part of the co-occurrence structure that the engineered closure features summarise, with the main hyperparameters and after re-tuning alike, whereas hybrids given those features came close to or matched gradient boosting and, for sparse rings, exceeded it. Message passing over two training periods with about 200 ring claims did not learn the shared-patient pattern that a single engineered feature states directly: at $s \geq 0.5$ the unsupervised overlap score alone (0.194 and 0.281) had higher ring AP than every pure GNN in both settings.

One caveat favours the engineered features: we designed them knowing the injected typology, a dense doctor–patient block. Their success shows that a few targeted features suffice when the typology is known; it does not show that they would capture typologies they were not designed for, where learned message passing could have an advantage. These findings are consistent with evidence from other domains that tree ensembles with neighbourhood features are strong graph anomaly detectors [10, 11, 32], with the same pattern on public Medicare provider data [12], and extend it to heterogeneous patient–provider claim graphs with controlled typologies.

## 9.3 Why a small fixed benchmark can point the other way

A small fixed benchmark can suggest conclusions that a controlled one does not support. We re-analysed one, a dataset of 2,100 claims with five injected rings produced by an earlier, simpler generator of ours (Online Resource 1, Section S10). It cannot tell whether relational structure helps: its ring claims were anomalous on their own (gradient boosting on three raw claim attributes reached a ring AP of 1.000), its rings recurred across the split, so every test ring claim was billed by a doctor with labelled ring claims in training, and its test split held only 21 ring claims. Controlling the feature-level signal, testing on new rings and replicating worlds removes all three confounds.

## 9.4 Public provider-labelled data

On the public provider-labelled data, gradient boosting on provider aggregates had the highest AP, and neither engineered graph features, label-neighbourhood features nor any GNN improved on it, in agreement with earlier comparisons on these data [12, 30]. This held over 25 rotations, where graph features changed GB-P's AP by at most

+0.006 to +0.007 (upper 95% limits), and with every model's configuration selected inside each rotation. The labels are not assortative through shared beneficiaries, and fraudulent providers differ mainly in volume, so the data sit in the regime where our benchmark also found no benefit from structure. This is consistent with the controlled results but is not an independent confirmation of the dense-ring regime: validating that regime requires real data with labelled collusion, which are not publicly available. Because the source of the claims and of the labels is undocumented, the result should also not be read as an estimate of how these methods perform on current Medicare claims.

## 9.5 Practical recommendations

- Start with gradient boosting on claim features, per-entity aggregates and a small set of shared-patient graph features. On our benchmark no pure GNN was better on average, its one significant per-regime advantage (0.005 AP) came where every method exceeded 0.98, and gradient boosting was 9–50 times cheaper to train than the GNNs and hybrids. Adding message passing on top of the engineered features gained about 0.01 AP, most for sparse rings; where rings may be sparse, a hybrid is worth testing, although every method detected sparse rings poorly when their members were not unusually active.
- Do not infer the value of relational structure from rings whose members are unusually active: then aggregates already find them. When they are not, explicit co-occurrence features, or hybrids that receive them, were needed.
- Evaluate on rings (or providers) that do not appear in training, and compare against a no-graph control with identical inputs; otherwise memorisation of known members can masquerade as relational learning. Report uncertainty over the independent units—generated worlds that share data, or cross-validation rotations that share training data, are not independent.
- If a GNN is used, include reverse relations, verify that information reaches the classified nodes, and give attention-based architectures structural (degree) encodings or prefer sum aggregation.
- Keep isolated-fraud detection on a claim-level model; message passing degraded it consistently.
- Do not expect a GNN second stage to pay for itself. Letting a GNN re-rank the claims that gradient boosting flags gained nothing here; averaging the two models' rankings gained at most 0.006 AP, and in an audit setting, where a whole period is scored at once, routing saves little computation (Section 6).

For a payer's health information system these results favour a modest design. The aggregates and shared-patient statistics used here are counts and overlaps over one audit period of claims, which a claims database can compute in batch, and the resulting model scored a whole period in a fraction of a second on a laptop CPU (Section 6). A GNN adds a graph-learning stack and 13–50 times the training cost of gradient boosting in our runs; a pure GNN brought no gain on average in our benchmark or on the public data, and a hybrid about 0.01 AP, most for sparse rings.

Because investigators examine only the highest-ranked cases, such systems should be evaluated at the investigation budget and on providers and rings unseen in training.

### 9.6 Limitations

The controlled findings hold for our generator. Its distributions are simple, labels are noise-free and immediate, rings are static during their active period and of a single size, and ring prevalence (8% of the non-isolated test-period claims at default volume) is far above realistic levels so that each world has enough positives for stable estimates; the ring-volume experiments lower it to 1.1–4.2% but cover only $\delta = 0$ and six seeds. The engineered graph features were designed with knowledge of the injected typology, which favours them over learned message passing (Section 9). We tuned each method once, on development worlds from a single regime ($\delta = 0.5$, $s = 0.5$), and once more for volume 4; per-regime tuning could change small differences. The twelve seeds per regime are the unit of replication: pooled tests have eleven degrees of freedom and cannot resolve differences much below 0.005–0.01 AP, and the persistent-ring, ablation, routing and ring-volume analyses rest on six seeds. Seeds 7–12 were added after the first results had been seen, under a plan fixed before the new runs; the first-round results remain in the repository. Each model was trained once per world. Graph models were trained full-batch on graphs of about 11,000 nodes per period; larger graphs would need sampling-based training [40]. The architecture and reliance analyses were run in one regime ($\delta = 0$, $s = 0.9$) with one model seed per world, and permutation importance can understate a feature group whose information is duplicated in another; these analyses describe what the trained models use, not causal effects of the generator. Routing was evaluated with one first stage, two GNNs and a grid of six escalation fractions, and cost excludes the upkeep of aggregates, graph features and degree counts as claims arrive. On the public data, labels are per provider, the source of the claims and of the labels is undocumented, and the observation window is a single year. Even 25 overlapping rotations leave the corrected tests little power for differences of a few hundredths of AP, and the repeated rotations covered only three of the five GNNs and neither hybrid. Each model was trained once per rotation. Finally, the benchmark does not model adaptive fraudsters, temporal ring dynamics, which temporal graph models target [50], or realistic coding semantics such as those of synthetic health records [51].

## 10 Reproducibility

All results are produced by one released package from raw inputs. Every run writes a record containing the git commit, a hash of its configuration, the hashes of the generated claim and ground-truth tables, the library versions and the per-claim (or per-provider) test predictions; the tables and figures of this paper are generated from these records by a single script. Automated tests, run in continuous integration, check that (i) no ground-truth column reaches any feature matrix and features are invariant to permuting the labels; (ii) a period's features and graph are unchanged when claims of other periods are perturbed; (iii) emergent rings in different splits share no doctor or patient; (iv) at $\delta = 0$ ring claims are distributed like legitimate claims; (v) features,

graphs and labels stay aligned by claim identifier; and (vi) for every GNN, perturbing doctor features changes claim logits, so relational information actually reaches the classified nodes; (vii) a routed score depends only on the claim's own two scores and validation statistics; (viii) rewiring the test period leaves the features of other periods unchanged; (ix) scoring a claim on its $L$-hop computation subgraph reproduces its full-graph score; and (x) the regimes of a seed share their legitimate claims and ring members, the design that makes the seed the unit of pooled tests. A further check confirms that every table in this article equals the file generated from the run records. The original runs were made from a working tree that had not yet been committed, so their records do not pin the code version; retraining GB-CAG, SAGE and HGT with the released code on all 54 worlds reproduced their stored test predictions (Section 4.4), which ties the reported numbers to the released code. Records now also store a hash of the research sources at process start. A checklist mapping this study to the REFORMS recommendations [52] and a datasheet for the generator [53] are included in the repository.

# 11 Conclusion

We asked when relational structure helps healthcare fraud detection and answered it with a benchmark that varies the feature-level signal, density and volume of provider rings independently, tests on rings never seen in training, and checks the conclusions on public provider-labelled data. Structure helped most when rings formed moderately or densely linked doctor–patient blocks, and it mattered most when ring members were not otherwise unusual: then per-entity aggregates detected little, while a ring filing about one claim per member could not be found at all. Gradient boosting with a handful of engineered shared-patient features matched or outperformed five heterogeneous GNNs on average at a fraction of the cost, and at reduced ring volume it detected far more rings than any pure GNN; adding message passing on top of those features gained about 0.01 AP, most for sparse rings, and routing high-risk claims to a GNN gained little. GNNs generalised to new rings only with reverse relations, sufficient depth and counting ability, degraded isolated-fraud detection, and could memorise known ring members in ways that inflate conventional evaluations. A small fixed benchmark whose rings are anomalous in their features, recur across the split and contribute few test claims can suggest the opposite conclusion. On public provider-labelled data no graph-based method improved on tabular aggregates, as earlier work on these data also found. These conclusions rest on a simulator whose engineered features were designed for its typology. We recommend that graph models for claims fraud be justified against a graph-augmented tabular baseline and a no-graph control, on entities unseen in training, with uncertainty computed over independent units. The benchmark, code and tests are publicly available to support such comparisons; the most valuable next step is real data with labelled collusive schemes.

**Supplementary information.** Online Resource 1 (PDF) contains the selected hyperparameters, the two-stage routing evaluation, the full ring-volume dose-response, the confirmatory test of the hybrid result, the cost measurements, the single-factor ablations of HGT, the input-reliance analysis, figures of the mechanism ablations and

of new versus recurring rings, repeated cross-validation and per-rotation tuning on the public data, and the re-analysis of the small fixed benchmark. The repository contains the per-world results (including standard deviations, F1, ROC-AUC and recall at a fixed investigation budget for every regime and method), the tuning logs and the run records from which every table and figure is generated. The per-claim and per-provider test predictions (about 30 MB) are not tracked in the repository; the released code regenerates them.

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- **Funding.** The authors did not receive support from any organization for the submitted work.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable: the study uses synthetic data and a public dataset released under a CC0 licence, in which beneficiaries, providers and physicians are identified only by pseudonymous codes; no human participants were involved and no new data were collected.
- **Consent for publication.** Not applicable: the manuscript contains no individual person's data in any form; it uses synthetic data and a public dataset with pseudonymous identifiers only.
- **Data availability.** The synthetic benchmark, at every ring volume studied, is generated by the released code from fixed seeds. The public provider-labelled data are available from Kaggle [47] under CC0; file checksums are listed in the repository.
- **Code availability.** Code, configurations, tests and result records are available at https://github.com/akshay-hudev/Fraud-Guard (directory `research/`) under the MIT licence. An archived release with a DOI will be provided upon acceptance.
- **Author contribution.** H M Akshay: conceptualization, methodology, software, investigation, formal analysis, data curation, visualization and writing—original draft. Gaurav S B, Eshita Narayana and Harshita Srivastava: methodology, investigation, formal analysis and writing—original draft. Malini M Patil: supervision and writing—review and editing. All authors read and approved the final manuscript.


## Appendix A Generator parameters

**Table A1** Generator parameters (defaults in `research/synth/generator.py`).

| Parameter | Value |
|---|---|
| Patients / doctors / hospitals / regions / specialties | 6,000 / 300 / 40 / 5 / 8 |
| Audit periods | $4 \times 6$ months (train: 1–2, validation: 3, test: 4) |
| Legitimate claims per patient | Poisson(gamma rate; mean 3, shape 1.5) over 24 months |
| Share of legitimate claims with a regular doctor | 0.85 (1–3 regular doctors per patient, same region) |
| Claim amount | log-normal: specialty mean + $0.15\,(\text{procedures} - 1)$ + $0.8\,\text{inpatient}$, s.d. 0.6 |
| Isolated fraud | 3% of legitimate volume; feature shift 2.0 |
| Ring size | 4 doctors, 12 recruited patients (utilisation-weighted) |
| Ring volume | 16 claims per month during activity; monthly burst days (ring-volume experiments: 8, 4 and 2) |
| Rings per period | 4 (emergent: new members each period; persistent: same 4 rings) |
| Feature shift $\delta$ | 0, 1, 2 (development worlds: 0.5) |
| Structural density $s$ | 0.1, 0.5, 0.9 (development worlds: 0.5) |
| Worlds per regime | 12 (seeds 1–12; the regimes of a seed share its legitimate claims, isolated fraud and ring members; secondary analyses use seeds 1–6); confirmatory test: seeds 7–18 at volume 4; development worlds: seeds 901, 902 |

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